

MLB - R Markdown Example

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This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document.

Loading the MLB Dataset

Reading the data from the Web.

```
mlb <- read.csv("https://michael.hahsler.net/SMU/DS_Workshop_Intro_R/examples/MLB_cleaned.csv")
```

Inspect the first few data rows.

Raw R output:

```
head(mlb)
```

```
##   First.Name Last.Name Team      Position Height.inches. Weight.pounds.  Age
## 1      Jeff   Mathis  ANA      Catcher         72           180 23.92
## 2      Mike   Napoli  ANA      Catcher         72           205 25.33
## 3      Jose   Molina  ANA      Catcher         74           220 31.74
## 4     Howie  Kendrick ANA First Baseman         70           180 23.64
## 5     Kendry  Morales ANA First Baseman         73           220 23.70
## 6      Casey Kotchman ANA First Baseman         75           210 24.02
```

A fancy option:

```
knitr::kable(head(mlb))
```

First.Name	Last.Name	Team	Position	Height.inches.	Weight.pounds.	Age
Jeff	Mathis	ANA	Catcher	72	180	23.92
Mike	Napoli	ANA	Catcher	72	205	25.33
Jose	Molina	ANA	Catcher	74	220	31.74
Howie	Kendrick	ANA	First Baseman	70	180	23.64
Kendry	Morales	ANA	First Baseman	73	220	23.70
Casey	Kotchman	ANA	First Baseman	75	210	24.02

An interactive option:

```
DT::datatable(mlb)
```

```
## PhantomJS not found. You can install it with webshot::install_phantomjs(). If it is installed, please
```

Data summary

The dataset contains 1034 players. Here is a summary:

```
knitr::kable(summary(mlb))
```

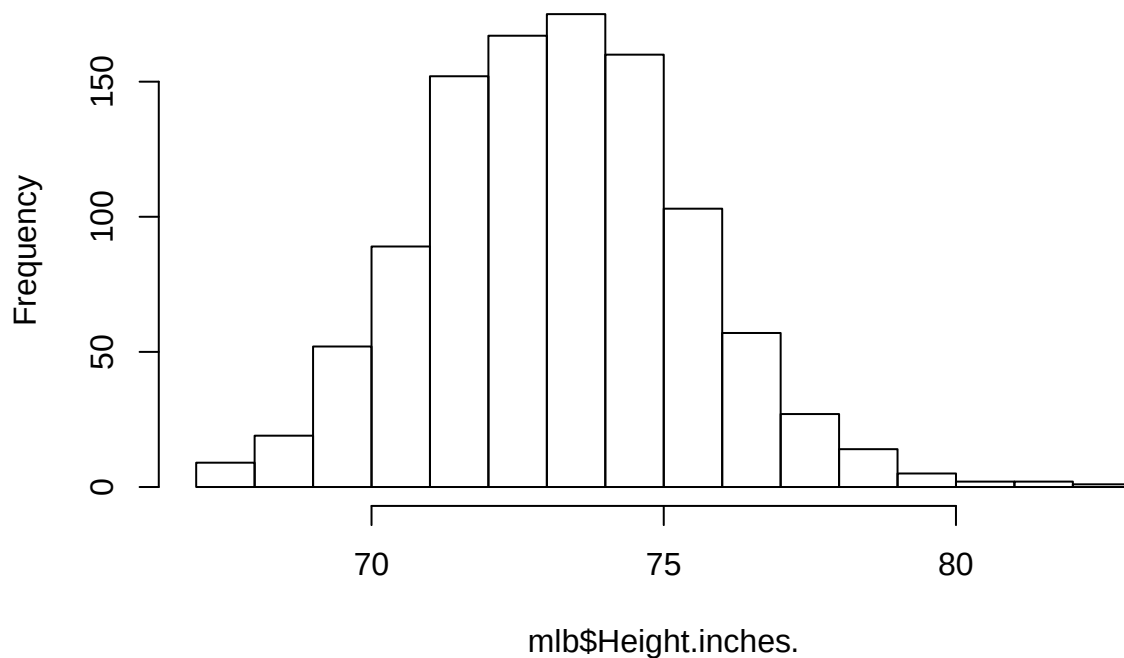
First.Name	Last.Name	Team	Position	Height.inches.	Weight.pounds.	Age
Jason : 27	Johnson : 9	NYM : 38	Relief Pitcher :315	Min. :67.0	Min. :150.0	Min. :20.90
Chris : 26	Perez : 7	ATL : 37	Starting Pitcher:221	1st Qu.:72.0	1st Qu.:187.0	1st Qu.:25.44
Mike : 26	Gonzalez : 6	DET : 37	Outfielder :194	Median :74.0	Median :200.0	Median :27.93
Scott : 24	Hernandez: 6	OAK : 37	Catcher : 76	Mean :73.7	Mean :201.7	Mean :28.74
Ryan : 23	Jones : 6	BOS : 36	Second Baseman : 58	3rd Qu.:75.0	3rd Qu.:215.0	3rd Qu.:31.23
Matt : 19	Ramirez : 6	CHC : 36	First Baseman : 55	Max. :83.0	Max. :290.0	Max. :48.52
(Other):889	(Other) :994	(Other):813	(Other) :115	NA	NA	NA

The height distribution

Base R:

```
hist(mlb$Height.inches.)
```

Histogram of mlb\$Height.inches.



Use plotly:

```
plotly::plot_ly(mlb, x = ~Height.inches., type = "histogram")
```

Exercises - Analyse an MLB Team

The goal of this exercise is to create a report that analyzes an MLB Team of your choice including basic information, tables and graphs.

1. Change the title and author.

2. Clean up the document and only keep the tables/figures you like.
3. Write what team you are analyzing.
4. Do a in-dept analysis of one team of your choice including tables and charts. Answer questions like:
 - How many players has the team in total?
 - Who is the youngest player?
 - How many players does the team have for each position?
 - What is the age range of the players?
 - Is there a difference in height and weight between positions?